Reproducibility and Explainability Analysis: Deep Learning for Customer Churn Prediction

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1 Introduction

Customer churn prediction is a critical problem in the telecommunications industry, as retaining existing customers is significantly more cost-effective than acquiring new ones. With the increasing availability of customer data, deep learning models have become a popular choice due to their ability to capture complex, non-linear relationships.

In this project, we aimed to reproduce a recent deep learning-based churn prediction study and evaluate its reproducibility. In addition, we extended the baseline by incorporating Explainable AI (XAI) techniques to better understand the model's decision-making process. The goal was not only to assess predictive performance but also to ensure that the model provides meaningful and interpretable insights.

2 Baseline Paper Overview

We selected the paper titled *"Highly Accurate Customer Churn Prediction in the Telecommunications Industry Using MLP"* (2024) as our baseline.

Objective

To develop a highly accurate churn prediction model using a Multilayer Perceptron (MLP).

Dataset

The study uses the IBM Telco Customer Churn dataset consisting of 7,043 customer records.

Methodology

The pipeline includes data preprocessing, class balancing using SMOTE, feature selection (removal of "gender"), and the application of an MLP model. The paper also mentions outlier detection using Tukey's Fences.

Reported Results

- Accuracy $\approx 92\%$
- AUC ≈ 0.98

These results indicate extremely high performance for a relatively small tabular dataset.

3 Implementation Methodology

Due to limited architectural details provided in the baseline paper, we implemented an MLP-based model inspired by the described approach, while following standard deep learning practices.

3.1 Data Pipeline

We replicated the preprocessing steps as closely as possible:

- Converted `TotalCharges` to numeric and removed missing values
- Dropped `customerID` and `gender` as per the paper
- Applied one-hot encoding to categorical variables

Standardization

Z-score normalization was applied to numerical features (`tenure`, `MonthlyCharges`, `TotalCharges`) using `StandardScaler`.

Class Imbalance Handling

SMOTE was used to balance the dataset.

Importantly, SMOTE was applied only to the training set after the train-test split to prevent data leakage.

3.2 Deep Learning Model

We implemented a deep neural network with the following structure:

$$\text{Input (29 features)} \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow \text{Output (1)}$$

Key Design Choices

- ReLU activation for non-linearity
- Batch Normalization and Dropout (0.3) for regularization
- Sigmoid activation at the output layer

Training Setup

- Loss Function: Binary Cross Entropy
- Optimizer: Adam (learning rate = 0.001)
- Batch Size: 64
- Maximum Epochs: 200
- Early Stopping with patience = 15

3.3 Experimental Setup

- Hardware: CPU-based training environment
- Data Split: 80% training, 20% testing
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score, and AUC

4 Experimental Results

Metric	**Baseline (2024)**	**Our Result**
Accuracy	92.00%	76.69%
AUC-ROC	0.98	0.8200
F1-Score	92.00%	0.5941
Recall	89.00%	0.6417

Our results are significantly lower than those reported in the baseline paper.

```
Loading data and model...
Loading dataset...
Splitting data (80% Train, 20% Test)...
Before SMOTE - Churners in train: 1495, Non-Churners: 4130
After SMOTE  - Churners in train: 4130, Non-Churners: 4130
Data preprocessing complete!

Calculating Final Test Metrics...
------------------------------
Accuracy:  0.7669
Precision: 0.5528
Recall:    0.6444
F1-Score:  0.5951
AUC:       0.8174
------------------------------
```

Figure 1: Training and Validation Loss Curves

5 Reproducibility Discussion

A key outcome of this project is that we were unable to reproduce the performance reported in the baseline paper.

5.1 Data Leakage Hypothesis

One likely explanation for this discrepancy is data leakage.

If SMOTE is applied before splitting the dataset, synthetic samples can introduce information from the training set into the test set. This leads to artificially inflated performance. In our implementation, we avoided this issue by applying SMOTE only to the training data.

As a result, our model's performance reflects a more realistic estimate of generalization on unseen data.

5.2 Model and Implementation Differences

The baseline paper does not provide sufficient architectural details. Therefore:

- We implemented a deeper MLP with regularization
- Used Batch Normalization and Dropout to improve generalization

These differences may also contribute to variation in results.

5.3 Missing Methodological Details

The paper mentions the use of Tukey's Fences for outlier detection but does not provide sufficient implementation details. Therefore, this step was not included in our reproduction.

5.4 Training Behavior

Our loss curves show that:

- Training loss decreases steadily
- Validation loss stabilizes early

This indicates mild overfitting and suggests that achieving extremely high performance (e.g., AUC = 0.98) is unlikely under proper evaluation conditions.

Figure 2: Training and Validation Loss Curves

6 Explainable AI (SHAP Analysis)

To improve interpretability, we applied SHAP (SHapley Additive exPlanations) to analyze model predictions.

Key Insights

- **MonthlyCharges:** Higher values strongly push predictions toward churn

- **Fiber Optic Internet:** Customers using fiber optic services show a higher likelihood of churn
- **Contract Type:** Long-term contracts (e.g., two-year plans) reduce churn probability

These findings align with real-world expectations and demonstrate that the model is learning meaningful patterns.

7 Conclusion

In this project, we implemented an MLP-based churn prediction model inspired by the baseline study and evaluated its reproducibility.

Although we could not replicate the extremely high performance reported in the paper, our results provide a more realistic and scientifically valid assessment of model performance.

Furthermore, by incorporating SHAP-based explainability, we extended the baseline by making the model more transparent and interpretable. This transforms the model from a purely predictive system into a practical decision-support tool.

8 References

1. Rabih, R., Sun, W., & Jamal, K. W. (2024). *Highly Accurate Customer Churn Prediction in the Telecommunications Industry Using MLP*. International Journal of Integrated Science and Technology (IJIST).
   Paper Link